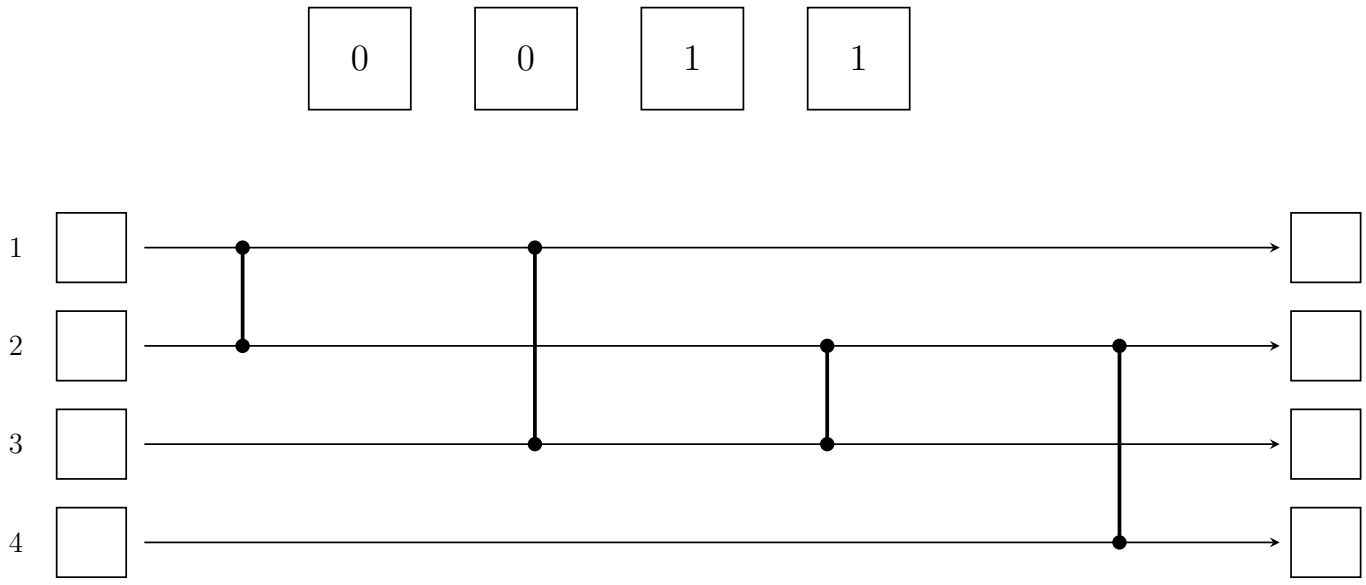
Move from left to right. At each bar, compare only the two lanes with black dots: the smaller value goes to the upper lane and the larger to the lower lane. Ties may stay put. Keep every bar fixed for every start; move cards only at bars. A machine sorts when the final values never get smaller from lane 1 down.

Problem 4: Use the cards 0, 0, 1, 1. Find every different starting order and test it on this machine. Decide whether the machine also sorts every other four-card start made of 0s and 1s.



Problem 5: Build a four-lane machine that sorts every start made of 0s and 1s. Make a complete record of the starting arrangements you checked.



Problem 6: Build a four-lane machine that sorts every order of 1, 2, 3, 4. Every bar must join neighboring lanes: 1 and 2, 2 and 3, or 3 and 4. Use as few bars as you can, and explain why fewer cannot work.

